# Implicit Differentiation

AP Calculus Worksheet · Grade 11–12

Name: ________________________________ Date: ____________________

**Learning Objectives**

- Distinguish between derivatives in terms of the same variable versus different variables
- Apply implicit differentiation to find dy/dx for equations involving x and y
- Solve for dy/dx algebraically after differentiating both sides of an equation

## Problems

**1.** Find the derivative of the function with respect to x using the power rule. Write your answer using prime notation.

$$f(x) = x^7$$

**2.** Find the derivative of y to the seventh power with respect to x using implicit differentiation. Keep the dy/dx notation in your answer.

$$y^7$$

**3.** Differentiate each term with respect to x. Keep the dy/dx notation for the y term.

$$x^4 + y^4$$

**4.** Use implicit differentiation to differentiate both sides of the equation with respect to x, then solve for dy/dx.

$$5y^9 = 1$$

**5.** Find dy/dx by differentiating both sides of the equation with respect to x using implicit differentiation.

$$x^2 + y^2 = 25$$

**6.** Differentiate both sides of the equation with respect to x and solve for dy/dx.

$$x^3 + y^3 = 8$$

**7.** Use implicit differentiation to find dy/dx for the equation below. Differentiate both sides with respect to x.

$$x^2y + y^3 = 10$$

**8.** Apply implicit differentiation to find dy/dx. Differentiate both sides with respect to x, then isolate dy/dx.

$$4x^2 - 3y^2 = 7x$$

**9.** Use implicit differentiation to find dy/dx for the equation below, which involves a product of x and y on the left side.

$$x^3y^2 - 5x + 2y = 0$$

**10.** Find the equation of the tangent line to the curve at the given point by first using implicit differentiation to find dy/dx, then substituting the point to find the slope.

$$x^2 + xy + y^2 = 7 \text{ at the point } (1, 2)$$

# Implicit Differentiation — Answer Key

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## Answer Key

**1. Answer: $f'(x) = 7x^6$**

- Apply the power rule: bring down the exponent and reduce it by 1.
- $f'(x) = 7x^{(7-1)} = 7x^6$

**2. Answer: $7y^6 \cdot dy/dx$**

- Apply the power rule to $y^7$, treating y as a function of x.
- The derivative is $7y^6$, but because we are differentiating y with respect to x, we must multiply by $dy/dx$.
- Final answer: $7y^6 \cdot dy/dx$

**3. Answer: $4x^3 + 4y^3 \cdot dy/dx$**

- Differentiate $x^4$ with respect to x: $4x^3$ (the $dx/dx = 1$ term drops out).
- Differentiate $y^4$ with respect to x using implicit differentiation: $4y^3 \cdot dy/dx$.
- Combine: $4x^3 + 4y^3 \cdot dy/dx$

**4. Answer: $dy/dx = 0$**

- Differentiate the left side: $d/dx(5y^9) = 45y^8 \cdot dy/dx$.
- Differentiate the right side: $d/dx(1) = 0$.
- Set them equal: $45y^8 \cdot dy/dx = 0$.
- Divide both sides by $45y^8$: $dy/dx = 0$.

**5. Answer: $dy/dx = -x/y$**

- Differentiate both sides with respect to x: $2x + 2y \cdot dy/dx = 0$.
- Isolate the dy/dx term: $2y \cdot dy/dx = -2x$.
- Divide both sides by 2y: $dy/dx = -x/y$.

**6. Answer: $dy/dx = -x^2/y^2$**

- Differentiate both sides: $3x^2 + 3y^2 \cdot dy/dx = 0$.
- Isolate dy/dx: $3y^2 \cdot dy/dx = -3x^2$.
- Divide both sides by $3y^2$: $dy/dx = -x^2/y^2$.

**7. Answer: $dy/dx = -2xy / (x^2 + 3y^2)$**

- Differentiate $x^2 \cdot y$ using the product rule: $2x \cdot y + x^2 \cdot (dy/dx)$.
- Differentiate $y^3$: $3y^2 \cdot dy/dx$.
- Differentiate 10: 0.
- Equation: $2xy + x^2 \cdot (dy/dx) + 3y^2 \cdot (dy/dx) = 0$.
- Factor dy/dx: $dy/dx(x^2 + 3y^2) = -2xy$.
- Solve: $dy/dx = -2xy / (x^2 + 3y^2)$.

**8. Answer: $dy/dx = (8x - 7) / (6y)$**

- Differentiate both sides: $8x - 6y \cdot dy/dx = 7$.
- Isolate the dy/dx term: $-6y \cdot dy/dx = 7 - 8x$.
- Divide both sides by -6y: $dy/dx = (8x - 7) / (6y)$.

**9. Answer: dy/dx = (5 - 3x^2·y^2) / (2x^3·y + 2)**

- Differentiate $x^3 \cdot y^2$ using the product rule: $3x^2 \cdot y^2 + x^3 \cdot (2y) \cdot (dy/dx)$.
- Differentiate $-5x$: $-5$.
- Differentiate $2y$: $2 \cdot (dy/dx)$.
- Full equation: $3x^2 \cdot y^2 + 2x^3 \cdot y \cdot (dy/dx) - 5 + 2 \cdot (dy/dx) = 0$.
- Group dy/dx terms: $dy/dx(2x^3 \cdot y + 2) = 5 - 3x^2 \cdot y^2$.
- Solve: $dy/dx = (5 - 3x^2 \cdot y^2) / (2x^3 \cdot y + 2)$.

**10. Answer: y = -(4/5)(x - 1) + 2, or y = -4x/5 + 14/5**

- Differentiate both sides with respect to x: $2x + (y + x \cdot dy/dx) + 2y \cdot (dy/dx) = 0$.
- Expand: $2x + y + x \cdot (dy/dx) + 2y \cdot (dy/dx) = 0$.
- Factor dy/dx: $dy/dx(x + 2y) = -(2x + y)$.
- Solve: $dy/dx = -(2x + y) / (x + 2y)$.
- Substitute (1, 2): $dy/dx = -(2 \cdot 1 + 2) / (1 + 2 \cdot 2) = -4/5$.
- Use point-slope form: $y - 2 = -4/5 \cdot (x - 1)$, so $y = -4x/5 + 14/5$.